

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac HelloWorld.java  
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java HelloWorld  
Hello, World!
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac Calculator.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java Calculator
```

```
Enter first number: 50
```

```
Enter second number: 48
```

```
Choose operation (+, -, *, /):
```

```
+
```

```
Result = 98.0
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac EvenOddChecker.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java EvenOddChecker
```

```
Enter an integer: 5
```

```
5 is Odd.
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac LeapYearChecker.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java LeapYearChecker
```

```
Enter a year: 2000
```

```
2000 is a Leap Year.
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac MultiplicationTable.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java MultiplicationTable
```

```
Enter a number: 5
```

```
Multiplication Table of 5
```

```
5 x 1 = 5
```

```
5 x 2 = 10
```

```
5 x 3 = 15
```

```
5 x 4 = 20
```

```
5 x 5 = 25
```

```
5 x 6 = 30
```

```
5 x 7 = 35
```

```
5 x 8 = 40
```

```
5 x 9 = 45
```

```
5 x 10 = 50
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac DataTypeDemo.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java DataTypeDemo
```

```
Integer value: 20
```

```
Float value: 5.8
```

```
Double value: 45000.75
```

```
Character value: A
```

```
Boolean value: true
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac TypeCastingExample.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java TypeCastingExample
```

```
Double value: 25.89
```

```
After casting to int: 25
```

```
Integer value: 50
```

```
After casting to double: 50.0
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac OperatorPrecedence.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java OperatorPrecedence
```

```
10 + 5 * 2 = 20
```

```
(10 + 5) * 2 = 30
```

```
20 / 4 + 3 * 2 = 11
```

```
8 + 4 / 2 * 3 = 14
```



```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac GradeCalculator.java
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java GradeCalculator
Enter marks out of 100: 98
Grade: A
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac NumberGuessingGame.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java NumberGuessingGame
```

```
Guess a number between 1 and 100
```

```
Enter your guess: 95
```

```
Too High!
```

```
Enter your guess: 52
```

```
Too High!
```

```
Enter your guess: 5
```

```
Too Low!
```

```
Enter your guess: 20
```

```
Too Low!
```

```
Enter your guess: 25
```

```
Too Low!
```

```
Enter your guess: 30
```

```
Too Low!
```

```
Enter your guess: 35
```

```
Too Low!
```

```
Enter your guess: 40
```

```
Congratulations! You guessed the correct number.
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac FactorialCalculator.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java FactorialCalculator
```

```
Enter a non-negative integer: 5
```

```
Factorial of 5 is 120
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac MethodOverloading.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java MethodOverloading  
Sum of two integers: 30  
Sum of two doubles: 31.0  
Sum of three integers: 60
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac MethodOverloading.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java MethodOverloading  
Sum of two integers: 30  
Sum of two doubles: 31.0  
Sum of three integers: 60
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac RecursiveFibonacci.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java RecursiveFibonacci
```

```
Enter a positive integer: 6
```

```
Fibonacci number at position 6 is 8
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac ArraySumAverage.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java ArraySumAverage
```

```
Enter number of elements: 5
```

```
Enter array elements:
```

```
1
```

```
2
```

```
3
```

```
5
```

```
6
```

```
Sum = 17
```

```
Average = 3.4
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac StringReverse.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java StringReverse
```

```
Enter a string: cognizant
```

```
Reversed String: tnazingoc
```



```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac PalindromeChecker.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java PalindromeChecker
```

```
Enter a string: cognizant
```

```
The string is not a palindrome.
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac CarDemo.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java CarDemo
```

```
Make: Toyota  
Model: Corolla  
Year: 2022
```

```
Make: Honda  
Model: Civic  
Year: 2023
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac InheritanceDemo.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java InheritanceDemo  
Animal makes a sound  
Bark
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac InterfaceDemo.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java InterfaceDemo  
Playing Guitar  
Playing Piano
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac TryCatchDemo.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java TryCatchDemo
```

```
Enter first integer: 5
```

```
Enter second integer: 8
```

```
Result = 0
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac CustomExceptionDemo.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java CustomExceptionDemo  
Enter your age: 20  
You are eligible.
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>javac FileWritingDemo.java  
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>java FileWritingDemo  
Enter text: cognizant  
Data written successfully to output.txt
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\23>javac FileReadingDemo.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\23>java FileReadingDemo  
File not found.
```



```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\24>javac ArrayListDemo.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\24>java ArrayListDemo
```

```
How many names do you want to add? 5
```

```
Enter name 1: cognizant
```

```
Enter name 2: amazon
```

```
Enter name 3: flipkart
```

```
Enter name 4: visa
```

```
Enter name 5: cisco
```

```
Student Names:
```

```
cognizant
```

```
amazon
```

```
flipkart
```

```
visa
```

```
cisco
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\25>javac HashMapDemo.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\25>java HashMapDemo
```

```
How many students? 4
```

```
Enter Student ID: 1
```

```
Enter Student Name: john
```

```
Enter Student ID: 2
```

```
Enter Student Name: rani
```

```
Enter Student ID: 3
```

```
Enter Student Name: raju
```

```
Enter Student ID: 4
```

```
Enter Student Name: rohan
```

```
Enter ID to search: 23
```

```
Student ID not found.
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\26>javac ThreadDemo.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\26>java ThreadDemo
```

```
Thread-1 is running
```

```
Thread-2 is running
```

```
Thread-1 is running
```

```
Thread-1 is running
```

```
Thread-1 is running
```

```
Thread-2 is running
```

```
Thread-1 is running
```

```
Thread-2 is running
```

```
Thread-2 is running
```

```
Thread-2 is running
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\27>javac LambdaSortDemo.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\27>java LambdaSortDemo
```

```
Sorted List:
```

```
Anjali
```

```
Bhanu
```

```
Kiran
```

```
Ravi
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\28>javac StreamDemo.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\28>java StreamDemo  
Even Numbers: [2, 4, 6, 8, 10]
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\29>javac RecordDemo.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\29>java RecordDemo
```

```
Person[name=Ravi, age=20]
```

```
Person[name=Anjali, age=17]
```

```
People age 18 and above:
```

```
Person[name=Ravi, age=20]
```

```
Person[name=Kiran, age=25]
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\30>javac PatternMatchingSwitch.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\30>java PatternMatchingSwitch
```

```
Integer value: 100
```

```
String value: Hello Java
```

```
Double value: 25.5
```

```
Boolean value: true
```

```
Object is null
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\31>javac -cp "lib/*" src/JDBCSelectDemo.java  
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\31>java -cp "lib/*;src" JDBCSelectDemo  
1 Ravi 20  
2 Anjali 21
```



```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\32>javac -cp "lib/*" src/*.java
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\32>java -cp "lib/*;src" Main
1 record inserted.
1 record updated.
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\33>javac -cp "lib/*" src/TransactionDemo.java  
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\33>java -cp "lib/*;src" TransactionDemo  
Transfer Successful
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>cd 34
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\34>javac -d mods/com.utils src/com.utils/module-info.java src/com.utils/com/utils/MessageUtil.java
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\34>javac --module-path mods -d mods/com.greetings src/com.greetings/module-info.java src/com.greetings/com/greetings/Main.java
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\34>java --module-path mods -m com.greetings/com.greetings.Main
Hello from Utility Module
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>cd 35
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\35>javac src/Server.java
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\35>javac src/Client.java
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\35>java -cp src Server
Server started...
Waiting for client...
Client connected!
Client: Hello Server!
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\35>java -cp src Client  
Server: Hello Client!
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>cd 36
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\36>javac src/HttpClientDemo.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\36>java -cp src HttpClientDemo
Status Code: 200
```

Response Body:

```
{
  "login": "octocat",
  "id": 583231,
  "node_id": "MDQ6VXNlcjU4MzIzMQ==",
  "avatar_url": "https://avatars.githubusercontent.com/u/583231?v=4",
  "gravatar_id": "",
  "url": "https://api.github.com/users/octocat",
  "html_url": "https://github.com/octocat",
  "followers_url": "https://api.github.com/users/octocat/followers",
  "following_url": "https://api.github.com/users/octocat/following{/other_user}",
  "gists_url": "https://api.github.com/users/octocat/gists{/gist_id}",
  "starred_url": "https://api.github.com/users/octocat/starred{/owner}/{/repo}",
  "subscriptions_url": "https://api.github.com/users/octocat/subscriptions",
  "organizations_url": "https://api.github.com/users/octocat/orgs",
  "repos_url": "https://api.github.com/users/octocat/repos",
  "events_url": "https://api.github.com/users/octocat/events{/privacy}",
  "received_events_url": "https://api.github.com/users/octocat/received_events",
  "type": "User",
  "user_view_type": "public",
  "site_admin": false,
  "name": "The Octocat",
  "company": "@github",
  "blog": "https://github.blog",
  "location": "San Francisco",
  "email": null,
  "hireable": null,
  "bio": null,
  "twitter_username": null,
  "public_repos": 8,
  "public_gists": 8,
  "followers": 22814,
  "following": 9,
  "created_at": "2011-01-25T18:44:36Z",
  "updated_at": "2026-05-22T11:38:32Z"
}
```

```

C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\35>cd..
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>cd 37
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\37>javac BytecodeDemo.java
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\37>java BytecodeDemo
Hello Java
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\37>javap -c BytecodeDemo
Compiled from "BytecodeDemo.java"
public class BytecodeDemo {
    public BytecodeDemo();
        Code:
            0: aload_0
            1: invokespecial #1                  // Method java/lang/Object."<init>":()V
            4: return

    public void greet();
        Code:
            0: getstatic     #7                  // Field java/lang/System.out:Ljava/io/PrintStream;
            3: ldc           #13                 // String Hello Java
            5: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
            8: return

    public static void main(java.lang.String[]);
        Code:
            0: new           #21                 // class BytecodeDemo
            3: dup
            4: invokespecial #23                 // Method "<init>":()V
            7: astore_1
            8: aload_1
            9: invokevirtual #24                 // Method greet:()V
            12: return
}

```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>cd 38
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\38>javac Hello.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\38>java Hello  
Hello Java
```


Hello.java

[Delete](#)

Hello.java

Download file

```
public class Hello {  
    public void display() {  
        System.out.println("Hello Java");  
    }  
  
    public static void main(String[] var0) {  
        Hello var1 = new Hello();  
        var1.display();  
    }  
}
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>cd 39
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\39>javac src/*.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\39>java -cp src ReflectionDemo  
Class Name: Student
```

```
Methods:
```

```
Method Name: display
```

```
Method Name: greet
```

```
Parameter Type: String
```

```
Hello from Student Class
```

```
Hello Pranavi
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\40>javac VirtualThreadDemo.java
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\40>java VirtualThreadDemo
```

```
Virtual Thread 1 is running  
Virtual Thread 66 is running  
Virtual Thread 101 is running  
Virtual Thread 139 is running  
Virtual Thread 766 is running  
Virtual Thread 8 is running  
Virtual Thread 6 is running  
Virtual Thread 797 is running  
Virtual Thread 808 is running  
Virtual Thread 824 is running  
Virtual Thread 834 is running  
Virtual Thread 847 is running  
Virtual Thread 4 is running  
Virtual Thread 852 is running  
Virtual Thread 864 is running  
Virtual Thread 873 is running  
Virtual Thread 880 is running  
Virtual Thread 893 is running  
Virtual Thread 901 is running  
Virtual Thread 906 is running  
Virtual Thread 919 is running  
Virtual Thread 926 is running  
Virtual Thread 934 is running  
Virtual Thread 943 is running  
Virtual Thread 950 is running  
Virtual Thread 959 is running  
Virtual Thread 970 is running  
Virtual Thread 983 is running  
Virtual Thread 992 is running  
Virtual Thread 999 is running  
Virtual Thread 1007 is running  
Virtual Thread 1016 is running  
Virtual Thread 1021 is running  
Virtual Thread 1067 is running  
Virtual Thread 1102 is running
```

```
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3>cd 41
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\41>javac ExecutorCallableDemo.java
C:\Users\User\OneDrive\Desktop\cognizant nurture 5.0\module_3\41>java ExecutorCallableDemo
Result 1: 30
Result 2: 70
Result 3: 110
```